1. Lesson 4: Distribution Theory and the Free Particle Problem in the Context of Superoscillations

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1.1. **Theory of distributions.** The theory of distributions was born to answer to different problems that lead to extend the notion of derivative.

Definition 1.1 (Test functions)**.** Let $\Omega \subseteq \mathbb{R}^n$. The space of test functions on Ω is the vectorial space

$$\mathcal{D}(\Omega) = C_0^\infty(\mathbb{R}^n).$$

Thus $\mathcal{D}(\Omega)$ is the space of function infinitely derivable in Ω with compat support K (where K is a closed and bounded set) in Ω.

An easy example of function test is given by

$$\varphi(x) = \begin{cases} e^{\frac{1}{x^2-1}}, & |x| < 1 \\ 0, & |x| \geq 1. \end{cases}$$

Now, we give the notion of convergence in the space $\mathcal{D}(\Omega)$.

Definition 1.2. Given a squence $\{\varphi_j\}_{j\in\mathbb{N}} \subset \mathcal{D}(\Omega)$ and let $\varphi \in \mathcal{D}(\Omega)$. We say that

$$\varphi_j \to \varphi \quad \text{in} \quad \mathcal{D}(\Omega).$$

If there exists a compact set $K \subset \Omega$ such that

- (i) $\text{supp}\varphi_j \subset K, \qquad \forall j$
- (ii) $\forall \alpha$ multi-index we have

$$\varphi_j \to \varphi \quad \text{and} \quad \partial^\alpha \varphi_j \to \partial^\alpha \varphi \qquad \text{uniformly in} \quad K.$$

Remark 1.3. In the one dimensional case: for any $k = 1, 2, 3, ...$

$$\varphi_j \to \varphi \quad \text{and} \quad \frac{d^k}{dx^k}\varphi_j \to \frac{d^k}{dx^k}\varphi \qquad \text{uniformly in} \quad K.$$

Now, we give the definition of set of distribution.

Definition 1.4. The set of distributions on Ω is denoted by $\mathcal{D}'(\Omega)$. It is the set all linear and continuos functionals:

$$T : \mathcal{D}(\Omega) \to \mathbb{R} \text{ (or } \mathbb{C}).$$

The notion of continuos functional is understood with respect to the convergence that we have introduced on $\mathcal{D}(\Omega)$. Precisely, let us consider $\{\varphi_j\}_{j\in\mathbb{N}} \in \mathcal{D}(\Omega)$, then T is continuos if

$$\varphi_j \overset{\mathcal{D}(\Omega)}{\to} \varphi \Rightarrow T(\varphi_j) \to T(\varphi).$$

Most of the times we will use the notations $\langle T, \varphi\rangle$ instead of $T(\varphi)$. This is to remark more the duality between $\mathcal{D}(\Omega)$ and $\mathcal{D}'(\Omega)$.
We observe that not all the functionals of $\mathcal{D}'(\Omega)$ can be represented as an integral with the Lebesgue measure. This is possible only in some cases.

Definition 1.5. The space of locally integrable functions on the open set $\Omega \subseteq \mathbb{R}^n$ is defined as

$$L^1_{loc}(\Omega) := \{f : \Omega \to \mathbb{C} \text{ measurable such that } f|_K \in L^1(K)\ \forall K \subset \Omega,\ K \text{ compact}\}.$$

Thus if we consider $f \in L^1_{loc}(\mathbb{R})$ we define the functional

$$T_f : \mathcal{D}(\Omega) \to \mathbb{R}$$

$$\langle T_f, \varphi \rangle := \int_{\mathbb{R}} f(x)\varphi(x)dx. \tag{1.1}$$

A well-known example of distribution that cannot be represented as an integral is given by the so called Dirac delta.

Definition 1.6. Let $x_0 \in \mathbb{R}$. We define the Dirac delta centred at the point $x_0 \in \mathbb{R}$ the functional

$$\delta_{x_0} : \mathcal{D}(\mathbb{R}) \to \mathbb{R}$$

$$\langle \delta_{x_0}, \varphi \rangle = \varphi(x_0).$$

Remark 1.7. We cannot write $\langle \delta_{x_0}, \varphi \rangle$ as $\int_{\mathbb{R}} \delta_{x_0}(x)\varphi(x)dx$. However the delta is characterized (with an abuse of notation) by the following property. For any continuos function f in $x = x_0$ for $a < b$ we have

$$\int_a^b f(x)\delta_{x_0}(x) = \begin{cases} f(x_0), & x_0 \in (a,b) \\ 0, & x_0 \notin (a,b) \end{cases} \tag{1.2}$$

Now, we give one of the most important notion in the theory of distributions.

Definition 1.8 (Derivative of a distribution)**.** For any $T \in \mathcal{D}'(\mathbb{R})$ we define the derivate of T the functional

$$T' : \mathcal{D}(\mathbb{R}) \to \mathbb{R}$$

$$\langle T', \varphi \rangle = -\langle T, \varphi' \rangle, \qquad \forall \varphi \in \mathcal{D}(\mathbb{R}).$$

Remark 1.9. One can prove that if T is a distribution, then also T' is a distribution as well.

Example 1.10. *Let*

$$H(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

We observe that $H \in L^1_{loc}(\mathbb{R})$. Thus by (1.1) *and integrating by parts, for any $\varphi \in \mathcal{D}(\mathbb{R})$, we get*

$$\begin{aligned} \langle T'_H, \varphi \rangle &= \int_{\mathbb{R}} [H(x)]'\varphi(x)dx \\ &= -\int_{\mathbb{R}} H(x)\varphi'(x)dx \\ &= -\int_0^{+\infty} \varphi'(x)dx \\ &= -\varphi(x)|_0^{+\infty} \\ &= -[\varphi(+\infty) - \varphi(0)] \\ &= \varphi(0) \\ &= \langle \delta_0, \varphi \rangle. \end{aligned}$$

Hence we can write

$$H'(x) = \delta_0(x) \quad in \quad \mathcal{D}'(\mathbb{R}).$$

Now, we want to introduce the Fourier transform of a distribution. We consider the particular case $T = T_f$ with $f \in L^1(\mathbb{R}^n)$. Let $\varphi \in \mathcal{D}(\mathbb{R}^n)$ we have

$$\langle T_{\hat{f}}, \varphi \rangle = \int_{\mathbb{R}^n} \hat{f}(y)\varphi(y)dy = {}^1 \int_{\mathbb{R}^n} f(y)\hat{\varphi}(y)dy = \langle T_f, \hat{\varphi} \rangle.$$

Hence the above computations suggest to have the following definition of the Fourier transform of $T \in \mathcal{D}'(\mathbb{R}^n)$:

$$\langle T_{\hat{f}}, \varphi \rangle = \langle T_f, \hat{\varphi} \rangle.$$

However in order to require that $\langle T_f, \hat{\varphi} \rangle$ makes sense we should have $\hat{\varphi} \in \mathcal{D}(\mathbb{R}^n)$. But this is a problem because it is not possible that both φ and $\hat{\varphi}$ have compact support. Thus in order to solve problem we have to consider the Fourier transform defined on the space of rapidly decreasing functions $\mathcal{S}(\mathbb{R}^n)$. This space is larger than $\mathcal{D}(\mathbb{R}^n)$ i.e. $\mathcal{D}(\mathbb{R}^n) \subseteq \mathcal{S}(\mathbb{R}^n)$. This follows from the following fact:

Lemma 1.11. *If $f \in C^\infty(\mathbb{R})$ and f has compact support then $f \in \mathcal{S}(\mathbb{R})$.*

Proof. We recall that the space of rapidly decreasing function is given by

$$\mathcal{S}(\mathbb{R}) := \{f \in C^\infty(\mathbb{R}) \,:\, \forall k, j \in \mathbb{N}_0 \,,\, \exists M_{k,j} \quad |t|^k |f^{(j)}(t)| \leq M_{k,j}\}.$$

If we suppose that $f(t)$ has compact support then it is zero outside the interval $[-T, T]$, with $T > 0$. Then we have

$$|t|^k |f^{(j)}(t)| = 0, \qquad |t| > T,$$
$$|t|^k |f^{(j)}(t)| \leq T^k \max_{t \in [-T,T]} |f^{(j)}(t)| = T^k M_j.$$

The last equality is justified (it means that the maximum exists and is finite) by the Weierstrass theorem there exists finite the $\max_{t\in[-T,T]} |f^{(j)}(t)|$ because $f^{(j)}(t)$ is continuos (since it is derivable). □

Remark 1.12. It is possible to consider a function $f \in \mathcal{S}(\mathbb{R})$ but it does not belong to the space $\mathcal{D}(\mathbb{R})$. We pick $f(t) = e^{-t^2}$. This function does not belong to $\mathcal{D}(\mathbb{R})$ since the support of this function is the real line. However $f(t) = e^{-t^2} \in \mathcal{S}(\mathbb{R})$. Indeed it is a C^∞ function and

$$f^(j)(t) = Q_j(t)e^{-t^2},$$

where $Q_j(t)$ is a polynomial coming from the derivation j-time of e^{-t^2}. Thus we have

$$|t|^k f^(j)(t) = |t|^k Q_j(t)e^{-t^2},$$

is bounded because $|t|^k Q_j(t)$ is a polynomial and the exponential e^{-t^2} grows faster then $|t|^k Q_j(t)$.

Thus we have that the corresponding set of linear and continuous functionals $\mathcal{S}'(\mathbb{R}^n)$ is properly contained in $\mathcal{D}'(\mathbb{R}^n)$. Thus the distributions in $\mathcal{S}'(\mathbb{R}^n)$ are particular distributions that are called *tempered distributions*. Now, we give the rigorous definition of this notion.

Definition 1.13. We say that T is a tempered distribution on $\mathbb{R}^n$, and we write $T \in \mathcal{S}'(\mathbb{R}^n)$, if T is linear functional on $\mathcal{S}(\mathbb{R}^n)$, continuos with respect to the convergence in $\mathcal{S}(\mathbb{R}^n)$ [2]:

$$f_j \to f \,\text{in}\, \mathcal{S}(\mathbb{R}^n) \Rightarrow \langle T, f_j \rangle \to \langle T, f \rangle.$$

[1] If f, $\varphi \in L^1(\mathbb{R}^n)$

[2] We say that a sequence $\{f_j\}_{j\geq 1} \subset \mathcal{S}(\mathbb{R}^n)$ goes to 0 in $\mathcal{S}(\mathbb{R}^n)$ if for any $k \in \mathbb{N}_0$ and for any muli-index α we have

$$\lim_{j\to\infty} \sup_{\mathbb{R}^n} |x|^k |\partial^\alpha f_j(x)| = 0$$

Remark 1.14. If we consider a function f defined such that

$$|f(x)| \leq C(1+|x|)^N, \tag{1.3}$$

for a constant $C > 0$ and an integer $N > 0$, we have that $T = T_f \in \mathcal{S}'(\mathbb{R}^n)$.

Definition 1.15. For any tempered distribution $T \in \mathcal{S}'(\mathbb{R}^n)$ we define the Fourier transform in the following way

$$\langle \hat{T}, \varphi \rangle = \langle T, \hat{\varphi} \rangle, \qquad \forall \varphi \in \mathcal{S}(\mathbb{R}^n).$$

The inverse Fourier transform is defined by

$$\langle \mathcal{F}^{-1}T, \varphi \rangle = \langle T, \mathcal{F}^{-1}\varphi \rangle, \qquad \forall \varphi \in \mathcal{S}(\mathbb{R}^n).$$

Remark 1.16. The properties of the Fourier transform in $\mathcal{S}'(\mathbb{R}^n)$ are equal to the one we have discussed in Lesson 2. We just observe that the Fourier transform given in Definition 1.15 is continuos on $\mathcal{S}'(\mathbb{R}^n)$

Example 1.17. Compute the Fourier transform of the δ_0 in $\mathbb{R}^n$. For $\varphi \in \mathcal{S}(\mathbb{R}^n)$ we have

$$\langle \mathcal{F}\delta_0, \varphi \rangle = \langle \delta_0, \hat{\varphi} \rangle = \hat{\varphi}(0) = \int_{\mathbb{R}^n} \varphi(x)dx = \langle T_1, \varphi \rangle.$$

This means that

$$\hat{\delta_0} = 1.$$

Thus from the inversion theorem we have

$$\hat{1} = \delta_0.$$

Now, we can compute the Fourier transform of "functions" that are neither in $L^1(\mathbb{R}^n)$ nor in $L^2(\mathbb{R}^n)$, but are bounded as (1.3).

Example 1.18. *Compute the Fourier transform of $e^{2\pi ix}$ in $\mathbb{R}$. Let $\varphi \in \mathcal{S}(\mathbb{R})$, then we have*

$$\langle \mathcal{F}(e^{2\pi ix}), \varphi \rangle = \langle e^{2\pi ix}, \hat{\varphi} \rangle = \int_{\mathbb{R}} e^{2\pi ix}\hat{\varphi}(x)dx = \varphi(1) = \langle \delta_1, \varphi \rangle,$$

where in the last equality we have used the definition of the inversion Fourier transform for rapidly decreasing functions. Thus by the definition of the Dirac delta (see Definition 1.6) we have

$$\mathcal{F}(e^{2\pi ix}) = \delta_1(x).$$

Generally speaking for fixed $a \in \mathbb{R}$ we have

$$\mathcal{F}(e^{2\pi aix}) = \delta_a(x). \tag{1.4}$$

For the inversion formula of the Fourier transform we have

$$\mathcal{F}(\delta_a) = e^{2\pi aix}.$$

1.2. **The Free Particle Problem in the Context of Superoscillations.** In the past 50 years, quantum physicists have discovered, and experimentally demonstrated, a phenomenon which they termed superoscillations. From the mathecamaical point view superoscillations are

Definition 1.19. Let $x \in \mathbb{R}$, $n \in \mathbb{N}$ and $a > 1$. We call the superoscillating sequence

$$Y_n(x,a) = \left(\cos\left(\frac{x}{n}\right) + ia\sin\left(\frac{x}{n}\right)\right)^n = \sum_{j=0}^{n} C_j(n,a)e^{2\pi i\omega_j(n)x},$$

where $\omega_j(n) := 1 - \frac{2j}{n}$ and $C_j(n,a) = \binom{n}{j}\left(\frac{1+a}{2}\right)^{n-j}\left(\frac{1-a}{2}\right)^j$.

Let $Y_n(x,a)$ as above. Then for every $x \in \mathbb{R}$ we have

$$\lim_{n\to\infty} Y_n(x,a) = e^{2\pi i a x},$$

The term superoscillations comes from the fact that the frequencies satisfy $|1-2j/n| \leq 1$, however $Y_n(x,a) \to e^{2\pi i a x}$, for $a > 1$, and the convergence is uniform on the compact sets in $\mathbb{R}$.

Now, we consider the evolution of superoscillatory sequence for the Schrödinger equation when it is taken as initial value for the free particle. We observe that since the initial data are not in L^2 we need to follow a different approach.
We consider the Cauchy problem

$$i\frac{\partial \psi(x,t)}{\partial t} = -\frac{\partial^2}{\partial x^2}\psi(x,t), \qquad \psi(x,0) = Y_n(x).$$

Theorem 1.20. *The time evolution of the spatial superoscillating sequence $Y_n(x)$, is given by*

$$\psi(x,t) = \sum_{j=0}^{n} C_j(n,a) e^{2\pi i \omega_j(n) x} e^{-(2\pi\omega_j(n))^2 it}.$$

Proof. To solve the Schrödinger equation with $Y_n(x)$ as initial condition, we will work in the space of tempered distributions $\mathcal{S}'(\mathbb{R})$ and use a Fourier transform argument. Taking the Fourier transform of the Schrödinger equation we obtain

$$i\frac{d\hat{\psi}(\lambda,t)}{dt} = (4\pi)^2\lambda^2\hat{\psi}(\lambda,t),$$

and integrating we get

$$\hat{\psi}(\lambda,t) = C(\lambda)e^{-(2\pi\lambda)^2 it} \tag{1.5}$$

where the arbitrary function $C(\lambda)$ can be determined by the initial condition and therefore

$$C(\lambda) = \hat{\psi}(\lambda,0) = \sum_{j=0}^{n} C_j(n,a)\mathcal{F}\left(e^{2\pi i\omega_j(n)\lambda}\right).$$

By (1.4) we have

$$C(\lambda) = \sum_{j=0}^{n} C_j(n,a)\delta_{\omega_j(n)}(\lambda).$$

By plugging the above expression in (1.5) we have

$$\hat{\psi}(\lambda,t) = \sum_{j=0}^{n} C_j(n,a)\delta_{\omega_j(n)}(\lambda)e^{-(2\pi\lambda)^2 it}.$$

Taking now the inverse Fourier transform and by (1.2) we have

$$\begin{aligned}
\psi(x,t) &= \int_{\mathbb{R}} \left[\sum_{j=0}^{n} C_j(n,a)\delta_{\omega_j(n)}(\lambda)e^{-(2\pi\lambda)^2 it}\right] e^{2\pi i\lambda x} d\lambda \\
&= \sum_{j=0}^{n} C_j(n,a)\int_{\mathbb{R}} \delta_{\omega_j(n)}(\lambda)e^{2\pi i\lambda x}e^{-(2\pi\lambda)^2 it} d\lambda \\
&= \sum_{j=0}^{n} C_j(n,a)e^{2\pi i\omega_j(n)x}e^{-(2\pi\omega_j(n))^2 it},
\end{aligned}$$

and the stamente follows.

$\square$

By using sophisticated arguments one can prove that the function $\psi_n(x,t)$ exhibits, for all values of t, the same superocillatory behaviour by $Y_n(x)$, see *The mathematics of superoscillations* by Y. Aharonov, F. Colombo, I. Sabadini, D.C. Struppa, J. Tollaksen, Mem. Amer. Math. Soc., 1174, 247, (2017), v+107 (arXiv: 1511.01938)